Failure Analysis Report

FAIL-0460 — Performance Degradation

1. Failure Summary

Failure ID	FAIL-0460
Failure Type	Performance Degradation
Severity	Minor
Component	Sensor Manager
Application	APP-006 — XRPlatform
Build Version	5.5.1
Timestamp	2024-12-17 22:24:37
Environment	ENV-003 — Hub Lab
OS Version	Tizen 7.0
Network	WiFi-6
Resolution	Open

2. Test Case Reference

Test Case ID	TC-0230
Title	Verify end-to-end flow of Bluetooth reconnection from device discovery to logout...
Execution ID	EXEC-02298
Test Run	RUN-003 — Regression Cycle 3 — Build 4.3.2
Requirement	REQ-0027 — Performance — Battery Optimization

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete Bluetooth reconnection within the defined SLA.
Observed Behaviour	The system exhibited performance degradation when the session token has expired.

4. Error Details

Error Message:

```
PerformanceViolation: p95 latency=2488ms exceeds SLA of 519ms.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.Diagnostics.execute(line 151)
```

Log Reference	LOG-26272

5. Executed Steps at Time of Failure

1	Disconnect the wearable and wait 24 seconds.	The audit log records the event with correct metadata.

SYNTHETIC DATA — No real Samsung confidential information

2	Verify that the UI displays 'Connection successful'.	The performance counter logs a value below the SLA threshold.
3	Capture a screenshot and attach as evidence.	The audit log records the event with correct metadata.
4	Navigate to Settings > Privacy and tap 'Verify Identity'.	The UI transitions to the next screen within 3 seconds.
5	Enter valid credentials: username='testuser_050', password='[REDA...	The performance counter logs a value below the SLA threshold.
6	Navigate to Settings > Account and tap 'Reset'.	No crash or ANR is observed.

6. Root Cause Analysis

Buffer overflow in the media codec when processing H.265 streams above 4K resolution.

7. Linked Defect

Defect ID	BUG-0480
Title	Performance Degradation in Sensor Manager: PerformanceViolation: p95 latency=248...
Severity	Minor
Priority	Medium
Status	IN_PROGRESS
Root Cause	Buffer overflow in the media codec when processing H.265 streams above 4K resolu...
Resolution	—
Assigned To	ENG-067

8. Historical Similar Failures

HIST-0259	FAIL-0260	Performance Degradation	6.1.9	2021-03-21	1	Resolved
HIST-0459	FAIL-0460	Performance Degradation	6.4.9	2024-05-08	12	Resolved
HIST-0521	FAIL-0522	Performance Degradation	5.5.7	2023-07-14	5	Open

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

SYNTHETIC DATA — No real Samsung confidential information